

Search.io: Powers move into ecommerce search with Google Cloud

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ABOUT SEARCH.IO

With Google Cloud Search.io has the and scalability to ecommerce search offerings and the target growth to customers within of years.



About Search.io

Founded in 2015, Search.io provides data-driven, AI-powered technologies that enable businesses to create and implement custom search solutions for their websites, ecommerce stores, or applications. To date, Search.io has grown to support more than 1 billion requests per month from hundreds of customers around the world. By late 2020, the business had grown to 20+ people between Australia and the United States.

Industries: Technology

Google Cloud results

- Supports 500+ product updates per second to indexes for large ecommerce customers
- Enables business to effectively execute shift to ecommerce market
- Supports near-term goals of tripling customer numbers and achieving a fivefold increase in queries

Enable
engine
spend
their ti
produc
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custom
activiti

Location: Australia

(<https://cloud.google.com/datastore/>),

App Engine (<https://cloud.google.com/appengine/>)

(<https://cloud.google.com/kubernetes-engine/>),

Cloud Datastore

(<https://cloud.google.com/datastore/>)

(<https://cloud.google.com/products/operations>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

(<https://cloud.google.com/container-registry/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Operations

(<https://cloud.google.com/products/operations>)

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Cloud Build (<https://cloud.google.com/build/>)

Container Registry

(<https://cloud.google.com/container-registry/>)

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enable businesses to create and implement custom search solutions for their websites, ecommerce stores or applications. To date, Search.io has grown to support more than 1 billion requests per month from hundreds of customers around the world.

By late 2020, the business had grown to 20+ people. After providing search for millions of web pages for companies in industries ranging from gaming to aerospace and defense manufacturers, Search.io expanded its focus to the ecommerce space. “Our goal is to enable any organization to create smart search experiences,” says Jon Silvers, Marketing Director at Search.io.

“Google Cloud
has learned how
to scale services
massively and
distribute them
around the globe
for its own
requirements.
As its technical
teams release
new products
and explain why
they’ve made

specific
decisions, you
can see they're
brilliantly
thought out."

—Hamish Ogilvy, Founder,
Search.io

"We started working with a leading Australian ecommerce company and achieved great success with our technology," says Jon. "Ecommerce companies understand that search can drive site conversions. Return on investment (ROI) can be demonstrated through increases in revenue or margins—clear financial goals which are tied to the search experience."

Search.io initially launched its search and matching technologies directly onto a public cloud service. However, search or match query latency varied considerably and compromised customers' experiences. "We do a lot of heavy data reading of millions of data points for the most complex matching queries, and we still want to come back with answers in milliseconds, so we need consistent compute performance and minimal network latency," says Hamish Ogilvy, founder of Search.io.

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—David Howden, co-
founder & Chief Engineer,
Search.io

**Faster compute and network
performance**

Search.io benchmarked Google Cloud (<https://cloud.google.com/>) and found compute and network performance was faster and more reliable than its incumbent provider could deliver. The business also evaluated Google's track record in running its own infrastructure, including its containerization technologies.

"Google Cloud has learned how to scale services massively and distribute them around the globe for its own requirements," says Hamish. "In addition, as its technical teams release new products and explain why they have made specific decisions, you can see they are brilliantly thought out."

He cites the advanced design of Google Cloud load balancing and support for the Kubernetes ([/kubernetes](https://kubernetes.io/)) open source container orchestration system as features that met Search.io's requirements.

"Products like Google Kubernetes Engine would allow us to spend very little time managing infrastructure while enjoying best-in-class service and monitoring out of the box, freeing up our engineers to concentrate on our frontline products," says David Howden, co-founder and Chief Engineer at Search.io.

A staged migration

Search.io decided to conduct a staged migration to Google Cloud and moved most of its workloads across over 18 months. The business is using App Engine ([/appengine](https://appengine.google.com/)) for web application development

and hosting, [Datastore](#) (/datastore) to run a scalable NoSQL application database, [BigQuery](#) (/bigquery) for data analysis and reporting, and [Google Kubernetes Engine](#) (/kubernetes-engine) to provide managed application containerization.

Search.io also uses [Cloud Profiler](#) (/profiler/docs) to profile the performance of CPU or memory-intensive functions across its production applications and make quick improvements; [Cloud Build](#) (/cloud-build) to build, test, and deploy in a serverless Ci/CD environment; and [Container Registry](#) (/container-registry) to store and manage container images.

“Google Cloud
frees us up to
concentrate on
product
development
and customer-
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products
themselves.”

*—David Howden, co-
founder & Chief Engineer,
Search.io*

Supporting ecommerce search

The Google Cloud infrastructure delivers the scalability and agility to support the organization's push into ecommerce search. "The change to our business has also impacted our technology requirements," says David. "While website content is typically quite static, in the ecommerce context, products, stock levels, purchase, and other information changes constantly, and this must be reflected in search results."

As a result, read/write performance and availability have become considerably more important for the business. "For large ecommerce vendors, we are making over 500 product updates per second to indexes, so our infrastructure needs to handle that while also serving queries," says David. "In addition, for ecommerce websites, there are dollars associated with downtime, so high availability has become much more important. We also need to respond to search requests quickly, as adding just 50 to 100 milliseconds to a request can affect whether a user completes an ecommerce transaction."

With Google Cloud, Search.io is achieving availability of about 99.99%, ensuring the business can meet the demands of customers.

Having increased its engineering workforce and consequently its software development and deployment load, the business relies heavily on Google Cloud to minimize its administrative overheads. “Google Cloud frees us up to concentrate on product development and customer-facing work –we spend 90%+ of our engineering time on the products themselves,” says David.

With Google Cloud, Search.io is now well positioned to realize its ambitious expansion plans. “Our aim is to reach 10,000 customers in a couple of years, which would require considerable growth of our supporting infrastructure,” says Jon. “In the near term, we hope to triple our number of customers and grow the number of queries fivefold, and we are extremely confident that Google Cloud can help us achieve this.

“As well as the lack of overhead required to get ourselves running and achieve the reliability we need, Google Cloud is extremely engineer friendly,” he adds.